



A New Two-Parameter Half-Cauchy Modified Extended Exponential Distribution: Simulation, and Application to Fatigue Life Data

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Received: 10th June, 2026

Accepted: 12th August, 2026

Published: 15th September, 2026

ABSTRACT

Background: This study introduces a simplified two-parameter type of the Half-Cauchy Modified Extended Exponential (HCME) distribution derived by modifying the half-Cauchy distribution and exponential distribution, leading to a more flexible lifetime distribution. We describe its statistical properties, including the probability density function, hazard function, and quantiles.

Method: A stable numerical implementation is used to estimate the parameters by maximum likelihood. The model is applied to a classical fatigue life dataset comprising 101 aluminum coupons (Birnbaum & Saunders, 1969). A simulation study is used to examine the finite-sample performance of the maximum likelihood estimators.

Results: The new HCME model exhibits an excellent fit with a Kolmogorov–Smirnov p -value of 0.945 and outperforms the three-parameter HCEE, classical Gamma, Weibull, and Lognormal distributions based on the Akaike information criterion (AIC = 915.61). The simulation study indicates that the estimators approach their true values as the sample size increases.

Conclusion: The proposed HCME model is an important addition to lifetime modeling, especially when a fast-increasing hazard rate is anticipated, with potential applications in mechanical engineering, material science, and quality control.

Keywords: Half-Cauchy distribution; Exponential distribution; Maximum likelihood; Fatigue data; Lifetime analysis

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INTRODUCTION

The classical exponential distribution is one of the fundamental distributions in survival and reliability analysis. Due to its constant hazard rate, it is often too restrictive for modeling lifetime datasets. Many extensions of the classical exponential distribution have been proposed, such as the generalized exponential distribution (Gupta and Kundu, 1999), the exponentiated Weibull distribution (Mudholkar and Srivastava, 1993), and the extended exponential distribution (Joshi, 2015). Chaudhary et al. (Chaudhary et al., 2022a) introduced a three-parameter Half-Cauchy Extended Exponential (HCEE) distribution by combining the half-Cauchy generator with the extended exponential baseline. However, some models contain a redundant parameter because parameters independently appear as a ratio (Chaudhary et al., 2022a), leading to over-parameterization. In this study, we propose a parsimonious two-parameter version, denoted HCME (β, λ), which avoids redundancy while retaining flexibility and validity.

The exponential distribution has been modified and extended in various ways, including the beta exponential distribution (Nadarajah and Kotz, 2006), the Kumaraswamy exponential distribution (Cordeiro and de Castro, 2011), the inverse exponentiated exponential Poisson distribution (Telee and Kumar, 2023), the al-

pha power transformed extended exponential distribution (Almarashi et al., 2019), and the inverse exponentiated odd Lomax exponential distribution (Chaudhary et al., 2022b). In many cases, the half-Cauchy distribution (Shaw, 1995) has been taken as a generator for new families because it has a heavy tail. Chaudhary and Kumar (Chaudhary and Kumar, 2022) used it to define the half-Cauchy modified exponential distribution. Telee et al. (Telee et al., 2025) introduced the beta-power half-Cauchy distribution, reducing the number of parameters.

Let X follow the half-Cauchy distribution with CDF

$$R(x; \theta) = \frac{2}{\pi} \arctan\left(\frac{x}{\theta}\right), \quad x > 0, \theta > 0. \quad (1)$$

The corresponding PDF of the half-Cauchy distribution is

$$r(t) = \frac{2}{\pi} \left(\frac{\theta}{\theta^2 + t^2} \right), \quad t > 0, \theta > 0. \quad (2)$$

The generating family of distributions developed by Zografos and Balakrishnan (Zografos and Balakrishnan, 2009) has the CDF

$$F(x) = \int_0^{-\ln[1-G(x)]} r(t) dt, \quad (3)$$

where $G(x)$ is the CDF of the baseline distribution and $r(t)$ is the PDF of any distribution.

Using the PDF defined in Eq. 2 in Eq. 3 gives

$$F(x) = \frac{2}{\pi} \arctan \left(-\frac{1}{\theta} \ln[1 - G(x)] \right). \quad (4)$$

The corresponding PDF is

$$f(x) = \frac{2g(x)}{\pi\theta[1 - G(x)]} \left[1 + \left\{ -\frac{1}{\theta} \log[1 - G(x)] \right\}^2 \right]^{-1}. \quad (5)$$

Taking the CDF of the extended exponential distribution as the baseline CDF instead of $G(x)$ gives the three-parameter HCEE defined in Eq (6).

$$F(x; \theta, \gamma, \lambda) = \frac{2}{\pi} \arctan \left(\frac{\gamma x e^{-\lambda/x}}{\theta} \right), \quad (6)$$

$$x > 0, \theta > 0, \gamma > 0, \lambda > 0.$$

Noting that only the ratio of the parameters is identifiable, we define the reduced modified model with CDF $F(x) = \frac{2}{\pi} \arctan(\beta x e^{\lambda x})$, $\beta = \frac{\gamma}{\theta}$. In this article, we use this simpler form to produce a distribution with a rapidly increasing hazard, a property observed in many mechanical fatigue datasets.

In some cases, model formulation using the half-Cauchy distribution with an exponential-family model as the baseline results in redundant parameters, leading to over-parameterization. This can reduce efficiency and increase model complexity without improving fit. This study addresses this issue by proposing a two-parameter version that maintains flexibility while eliminating redundancy.

The important contributions of this study are:

- Derivation of a reduced two-parameter HCME distribution that exhibits a rapidly increasing hazard rate.
- Derivation and discussion of statistical properties, including moments and the quantile function.
- Demonstration of the fit of the proposed model on the fatigue dataset among eleven probability models.
- A simulation study examining the finite-sample behavior of the estimators.

The rest of the study is organized as follows. Section 2 explains the formulation of the new distribution and derives its main properties. Section 3 discusses maximum likelihood estimation and presents a simulation study. Section 4 applies the model to real fatigue data and compares it with several competing distributions. Section 5 concludes the study.

THE TWO-PARAMETER HCME DISTRIBUTION

Cumulative distribution and probability density functions

The cumulative distribution function (CDF) of the proposed HCME (β, λ) distribution is

$$F(x; \beta, \lambda) = \frac{2}{\pi} \arctan(\beta x e^{\lambda x}), \quad x > 0, \beta > 0, \lambda > 0. \quad (7)$$

The corresponding probability density function (PDF) is obtained by differentiation:

$$f(x; \beta, \lambda) = \frac{2\beta e^{\lambda x}(1 + \lambda x)}{\pi [1 + (\beta x e^{\lambda x})^2]^2}, \quad x > 0. \quad (8)$$

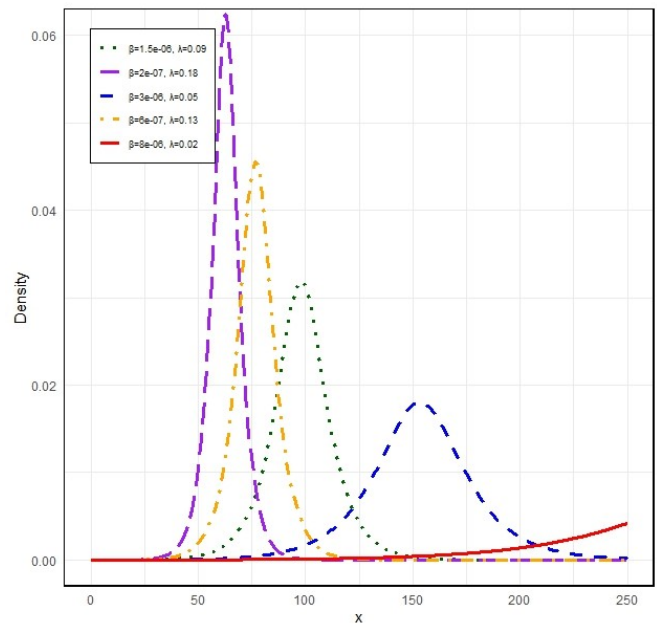


Figure 1. Density plots for five sets of parameters.

It is clear to verify that $F(0) = 0$ and $F(\infty) = 1$. The PDF has a single modal value for all parameter values and can exhibit a variety of shapes, ranging from nearly exponential to highly peaked (Figure 1). Similarly, the survival function is

$$S(x) = 1 - F(x; \beta, \lambda) = 1 - \frac{2}{\pi} \arctan(\beta x e^{\lambda x}), \quad x > 0, \beta > 0, \lambda > 0. \quad (9)$$

The hazard rate function is

$$h(x) = \frac{2}{\pi} \frac{\beta e^{\lambda x}(1 + \lambda x)}{\left[1 + (\beta x e^{\lambda x})^2 \right] \left[1 - \frac{2}{\pi} \arctan(\beta x e^{\lambda x}) \right]}, \quad x > 0. \quad (10)$$

The hazard rate is increasing, making the HCME model suitable for wear-out phenomena.

Quantile function and random generation

The quantile function $Q(u) = F^{-1}(u)$ does not have a closed form. It can be obtained numerically by solving

$$\beta x e^{\lambda x} = \tan\left(\frac{\pi u}{2}\right) \quad (11)$$

for x . Random variates can be generated using the inversion method.

Moments

For large x , moments of the HCME distribution are finite because the tail decays as

$$S(x) \sim \frac{2(1 + \lambda x)}{\pi \beta x e^{\lambda x}}, \quad x \rightarrow \infty, \quad (12)$$

and is therefore extremely light with exponential decay. Since the parameter β is often very small, the distribution can be highly concentrated; numerical integration is required to compute moments.

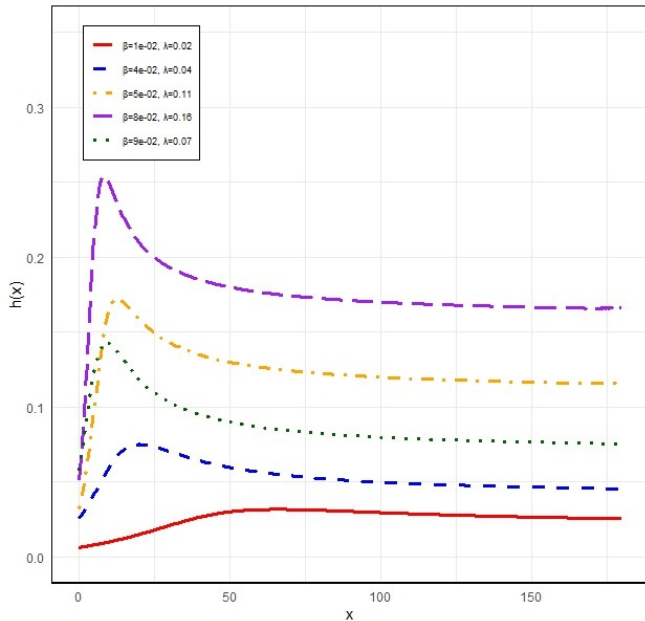


Figure 2. Hazard rate plots for five sets of parameters.

ESTIMATION AND SIMULATION

Maximum likelihood estimation

Let $X_1, \dots, X_n$ be an i.i.d. sample from $HCME(\beta, \lambda)$. The log-likelihood function is

$$\ell(\beta, \lambda) = n \ln \left(\frac{2}{\pi} \right) + n \ln \beta + \sum_{i=1}^n \ln(1 + \lambda x_i) + \lambda \sum_{i=1}^n x_i - \sum_{i=1}^n \ln [1 + \beta^2 e^{2\lambda x_i} x_i^2] \quad (13)$$

To ensure positivity, we reparameterize with $\theta_1 = \ln \beta$ and $\theta_2 = \ln \lambda$ and maximize ℓ using the BFGS algorithm. The Hessian matrix provides standard errors via the delta method. The first-order, second-order, and cross derivatives of Eq. 13 are used to obtain the estimated values of β and λ .

$$\frac{\partial \ell}{\partial \beta} = \frac{n}{\beta} - 2\beta \sum_{i=1}^n \frac{x_i^2 e^{2\lambda x_i}}{1 + \beta^2 x_i^2 e^{2\lambda x_i}}, \quad (14)$$

$$\frac{\partial \ell}{\partial \lambda} = \sum_{i=1}^n \left[\frac{x_i}{1 + \lambda x_i} + x_i - \frac{2\beta^2 x_i^3 e^{2\lambda x_i}}{1 + \beta^2 x_i^2 e^{2\lambda x_i}} \right], \quad (15)$$

$$\frac{\partial^2 \ell}{\partial \beta^2} = -\frac{n}{\beta^2} - 2 \sum_{i=1}^n \frac{x_i^2 e^{2\lambda x_i} (1 - \beta^2 x_i^2 e^{2\lambda x_i})}{(1 + \beta^2 x_i^2 e^{2\lambda x_i})^2}, \quad (16)$$

$$\frac{\partial^2 \ell}{\partial \beta \partial \lambda} = \frac{\partial^2 \ell}{\partial \lambda \partial \beta} = -4\beta \sum_{i=1}^n \frac{x_i^3 e^{2\lambda x_i}}{(1 + \beta^2 x_i^2 e^{2\lambda x_i})^2}, \quad (17)$$

$$\frac{\partial^2 \ell}{\partial \lambda^2} = -\sum_{i=1}^n \frac{x_i^2}{(1 + \lambda x_i)^2} - 4\beta^2 \sum_{i=1}^n \frac{x_i^4 e^{2\lambda x_i}}{(1 + \beta^2 x_i^2 e^{2\lambda x_i})^2}. \quad (18)$$

The Hessian matrix is

$$H(\beta, \lambda) = \begin{pmatrix} \frac{\partial^2 \ell}{\partial \beta^2} & \frac{\partial^2 \ell}{\partial \beta \partial \lambda} \\ \frac{\partial^2 \ell}{\partial \lambda \partial \beta} & \frac{\partial^2 \ell}{\partial \lambda^2} \end{pmatrix}. \quad (19)$$

At the maximum likelihood estimates, the observed information matrix is defined as

$$I(\hat{\beta}, \hat{\lambda}) = -H(\hat{\beta}, \hat{\lambda}). \quad (20)$$

The estimated variance-covariance matrix of the maximum likelihood estimators is then obtained from

$$\widehat{\text{Var}} \begin{pmatrix} \hat{\beta} \\ \hat{\lambda} \end{pmatrix} = I(\hat{\beta}, \hat{\lambda})^{-1}. \quad (21)$$

Consequently, the standard errors of $\hat{\beta}$ and $\hat{\lambda}$ are obtained as the square roots of the diagonal elements of this variance-covariance matrix. These standard errors can subsequently be used to construct confidence intervals and conduct statistical inference for the model parameters.

Under standard regularity conditions, the MLEs are consistent and asymptotically normal:

$$\sqrt{n} (\hat{\theta} - \theta) \xrightarrow{d} N(0, I^{-1}(\theta)), \quad (22)$$

where $I(\theta)$ is the Fisher information matrix. Using asymptotic normality, approximate 95% confidence intervals for β and λ are

$$\hat{\beta} \pm 1.96 SE(\hat{\beta}), \quad \hat{\lambda} \pm 1.96 SE(\hat{\lambda}). \quad (23)$$

Cramér-von Mises Estimation

The discrepancy between the fitted CDF and the empirical distribution is

$$W^2(\beta, \lambda) = \frac{1}{12n} + \sum_{i=1}^n \left[F(x_{(i)}; \beta, \lambda) - \frac{2i-1}{2n} \right]^2. \quad (24)$$

Minimizing Eq. 24 gives the Cramér-von Mises estimates.

Least Square Estimation

Least-square estimation (LSE) estimates the parameters by minimizing

$$S(\beta, \lambda) = \sum_{i=1}^n \left[F(x_{(i)}; \beta, \lambda) - \frac{i}{n+1} \right]^2. \quad (25)$$

Minimizing Eq. 25 gives the LSE estimates.

Simulation study

A Monte Carlo simulation was performed to evaluate the finite-sample performance of the MLEs. Samples of size $n = 100$ were generated from the HCME distribution using $\beta = 2.3487 \times 10^{-6}$ and $\lambda = 0.060539$, the values obtained from the real data. Over $R = 200$ replications, convergence of the MLEs occurred. The results are shown in Table 1.

The estimated values show some bias for small samples, whereas the root mean squared errors are acceptable. For larger sample sizes (e.g., $n = 500, 1000$), the bias becomes negligible.

Table 1. Monte Carlo simulation results for finite sample performance.

Parameter	True value	Mean estimate	Bias (relative)	RMSE (relative)
β	2.3487×10^{-6}	2.671×10^{-6}	+13.8%	100.3%
λ	0.06054	0.07642	+26.2%	28.4%

APPLICATION TO FATIGUE LIFE DATA

Data description

To test the applicability of the model, we consider the classical fatigue life dataset of 6061-T6 aluminum coupons tested at 18 cycles per second with maximum stress of 31,000 psi (Birnbaum and Saunders, 1969). There are 101 observations (in thousands of cycles) in the dataset:

70, 90, 96, 97, 99, 100, 103, 104, 104, 105, 107, 108, 108, 108, 109, 109, 112, 112, 113, 114, 114, 114, 116, 119, 120, 120, 120, 121, 121, 123, 124, 124, 124, 124, 124, 128, 128, 129, 129, 130, 130, 130, 131, 131, 131, 131, 131, 132, 132, 132, 133, 134, 134, 134, 134, 134, 136, 136, 137, 138, 138, 138, 139, 139, 141, 141, 142, 142, 142, 142, 142, 144, 144, 145, 146, 148, 148, 149, 151, 151, 152, 155, 156, 157, 157, 157, 157, 158, 159, 162, 163, 163, 164, 166, 166, 168, 170, 174, 196, 212.

All computations were performed using R version 4.3.1 (R Core Team, 2023) with the `optim` function for MLE and the `nleqslv` package (Hasselman and Carnell, 2026) for solving the likelihood equations. The fatigue life dataset is available in the NeuDist package (Kumar et al., 2026), which also provides diagnostic procedures including Q-Q plots, P-P plots, and goodness-of-fit tests. The fatigue life data exhibit the characteristics shown in Table 2. The positive skewness and heavy tail suggest that a distribution with increasing hazard is appropriate, motivating the use of the HCME model.

Table 2. Descriptive statistics of the fatigue life data.

n	Min	Mean	Median	SD	Variance
101	70	133.73	133	22.36	499.78
		Q1	Q3	Skewness	Kurtosis
		120	146	0.3305	4.052

Maximum likelihood results

Parameter estimation during modeling is an essential part of the analysis. There are various statistical tools for parameter estimation. In this study, we used maximum likelihood estimation (MLE). The MLEs, together with the LSE and CVM estimates, are shown in Table 3.

The estimated β around 2.35×10^{-6} and $\lambda = 0.0605$ describe the failure process. The small β value shows that the modification of half-Cauchy has a subtle but important effect on the tail behavior. Parameter λ determines the scale of the failure process. Using asymptotic normality, approximate 95% confidence intervals for β and

Table 3. Estimated parameters and standard error of estimates.

Estimation method	$\hat{\beta}$	$\hat{\lambda}$
MLE	2.35×10^{-6}	0.06054
LSE	3.33×10^{-6}	0.05793
CVM	3.34×10^{-6}	0.05796

λ are

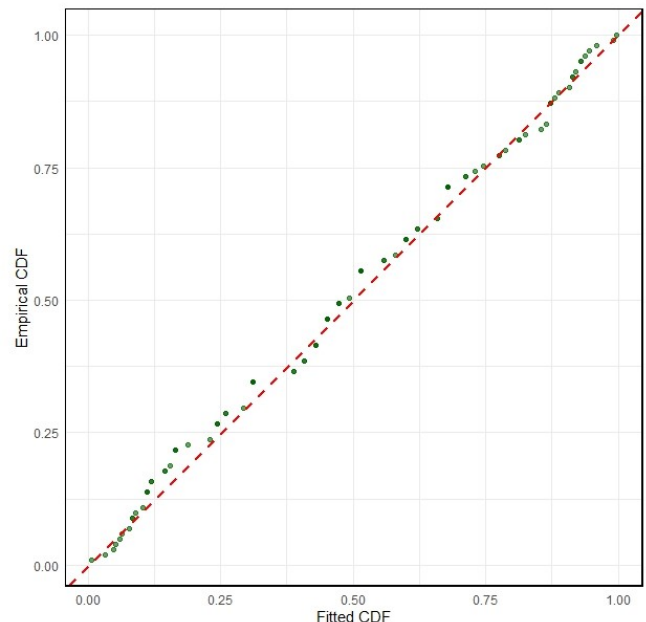
$$\beta : (1.98 \times 10^{-6}, 2.72 \times 10^{-6})$$

and

$$\lambda : (0.052, 0.069).$$

Model validation and Goodness-of-fit

To demonstrate that the proposed model HCME fits the dataset more precisely, P-P plot Figure 3 and Q-Q plot Figure 4 were plotted. Most of the data points fall around the straight line, confirming the suitability of the proposed model.

**Figure 3.** The P-P plot.

Although various techniques for testing the goodness-of-fit of the model are available, the Kolmogorov-Smirnov test is used in this study. The test statistic between the empirical CDF and the fitted HCME CDF is $D = 0.05227$ with a p -value of 0.945, indicating an excellent fit for the given dataset. In addition, the Anderson-Darling statistic is $AD = 0.284$ with $p > 0.05$, and the Cramér-von Mises statistic is $CVM = 0.042$ with $p > 0.05$; both

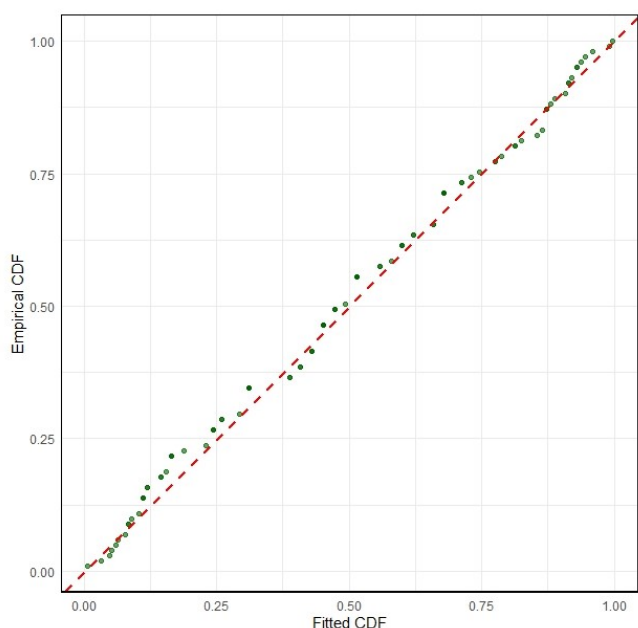


Figure 4. The Q-Q plot

confirm the excellent fit of the HCME model. Figure ?? displays the histogram with the fitted density and the empirical CDF against the fitted CDF; both confirm the adequacy of the model.

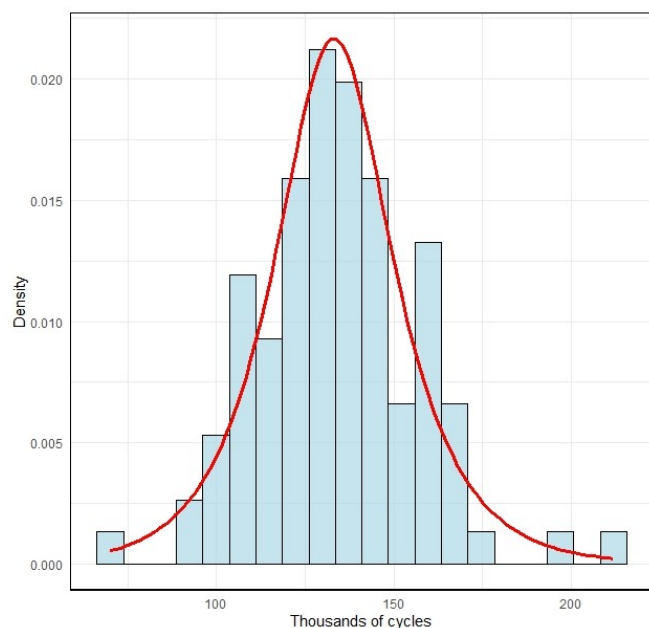


Figure 5. Histogram with the fitted density

Comparison with other distributions

We compared the HCME model with ten other probability models: the half-Cauchy extended exponential (HCEE) distribution (Chaudhary et al., 2022a), Gamma, Lognormal, Weibull, Chen distribution (Chen, 2000), Exponential Extension (EE) distribution (Nadarajah and Haghighi, 2011), Generalized Exponential (GE) distribution (Gupta and Kundu, 1999), Exponentiated Weibull

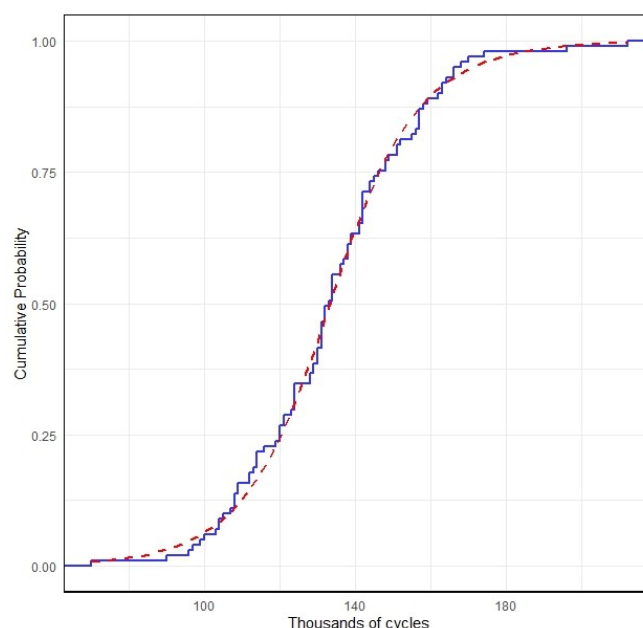


Figure 6. Plot of empirical CDF against the fitted CDF.

(EW) distribution (Mudholkar and Srivastava, 1993), Modified Weibull (MW) distribution, and Weibull Extension (WE) distribution (Tang et al., 2003). The AIC values are displayed in Table 4.

Table 4. Log-likelihood values, AIC, AIC weights, and KS p -values.

Distribution	Log-likelihood	AIC	AIC weight	KS p -value
HCME	-455.80	915.61	0.687	0.945
Lognormal	-457.12	918.24	0.178	0.055
Half-Cauchy Extended Exponential	-458.54	921.08	0.052	0.800
Weibull	-458.93	921.86	0.045	0.042
Exponentiated Weibull	-458.76	923.52	0.020	0.170
Gamma	-460.90	925.80	0.008	0.200
Exponential Extension	-460.90	925.79	0.008	0.096
Generalized Exponential	-463.73	931.46	0.001	0.201
Chen	-467.06	938.12	0.000	0.172
Weibull Extension	-466.00	938.01	0.000	0.000
Modified Weibull	-469.43	944.85	0.000	0.005

The proposed two-parameter HCME yields the lowest AIC among all compared models, demonstrating its superiority for this dataset. The Akaike weight of 0.687 for HCME indicates that it is the best-supported model among those considered according to the reported weights.

Limitations and Future Research

Despite the excellent performance of HCME, several limitations should be acknowledged:

- Censored data:** The current formulation assumes

complete observations. Extending the model to right-, left-, or interval-censored data would broaden its applicability in survival analysis.

- ii. **Bayesian inference:** Since the present study relies solely on maximum likelihood estimation, a Bayesian approach with informative priors could provide additional insights.
- iii. **Small-sample performance:** The simulation study focused on $n = 100$. Future work should investigate smaller sample sizes and higher censoring rates.
- iv. **Regression framework:** The current model is univariate. A regression version incorporating covariates would enable its use in accelerated life testing and reliability engineering.

CONCLUSION

We have introduced a two-parameter HCME distribution by modifying the half-Cauchy and exponential distributions, thereby overcoming the redundancy of the original three-parameter version. The model is mathematically well-defined with varying shapes of the density and hazard-rate plots. It is straightforward to estimate the parameters using maximum likelihood and exhibits an excellent fit to real fatigue life data. The HCME model achieved the smallest AIC and a high KS p -value compared with several classical and recent lifetime models. Future work includes deriving other important properties, extending the distribution to accommodate censored data, regression modeling, and applying other methods of parameter estimation, including Bayesian inference procedures.

FUNDING

This research is not funded by any external sources.

INSTITUTIONAL REVIEW BOARD STATEMENT

Not applicable.

CONSENT STATEMENT

Not applicable.

DATA AVAILABILITY STATEMENT

The data presented in this study are available in the manuscript.

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Citation: Sah Telee, L.B., & Yadav, U.K.(2026). A New Two-Parameter Half-Cauchy Modified Extended Exponential Distribution: Simulation, and Application to Fatigue Life Data. *Nepal Journal of Statistics and Data Sciences*. 1(1),50–56. DOI: [10.68031/njsd.v1i.00006](https://doi.org/10.68031/njsd.v1i.00006)